9.5 The Sine and Cosine Ratios

Name _____ Period ____ Date _____

Geometry 9.5 · Student sheet

I can…

- find the sine and cosine of an acute angle in a right triangle.
- use sine and cosine to find missing sides of a right triangle.
- use the cofunction relationship and angles of depression to solve problems.

Vocabulary · fill in the missing word

Sine — sin A = (leg opposite ∠A) ÷ _____

Cosine — cos A = (leg adjacent to ∠A) ÷ _____

Complementary angles — two angles whose _____ add to 90°

Angle of depression — the angle from the _____ down to your line of sight

Part 1 · Discovery: the other two ratios

Draw a right triangle with a **35°** angle and a hypotenuse of **10 cm**. Measure both legs. The other acute angle is 55°.

Angle	Opposite ÷ hypotenuse	Adjacent ÷ hypotenuse	sin on the calculator	cos on the calculator
35°				
55°				

What do you notice about the 35° row and the 55° row?

Part 2 · Notes and examples (fill in each note, then work the example)

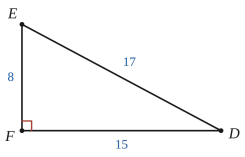
Sine: sin A = _____ ÷ hypotenuse

Cosine: cos A = _____ ÷ hypotenuse

SOH-CAH-TOA: Sine = Opposite ÷ _____, Cosine = _____ ÷ Hypotenuse, Tangent = Opposite ÷ _____

Example 1

Find sin D, cos D, sin E, and cos E. Write each as a fraction and as a decimal rounded to four places.



Cofunctions: if A and B are complementary, then sin A = _____ and cos A = _____.

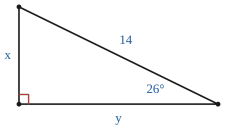
Example 2

a. Write cos 37° in terms of sine.

b. Write sin 20° in terms of cosine.

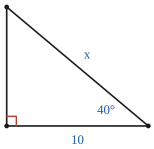
Example 3

Find the values of x and y. Round to the nearest tenth.



Example 4

Find the value of x. Round to the nearest tenth.



Angle of depression: the angle from the horizontal _____ to your line of sight. It is congruent to the angle of _____ from the other end.

Example 5

You are at the top of a cliff 120 feet above the water. The angle of depression to a boat is 18°. How far is the boat from you along your line of sight? Round to the nearest tenth.

Part 3 · Partner practice

Partner 1

A right triangle has legs of 7 and 24 and a hypotenuse of 25. ∠A is across from the leg of 7. Find sin A and cos A.

Partner 2

Find the value of x. Round to the nearest tenth.

